

Amardeep Chaubey

STAFF DATA ENGINEER

amar.xeon@gmail.com · London, UK · [linkedin.com/in/amardeepchaubey](https://www.linkedin.com/in/amardeepchaubey) · github.com/xeonee

PROFESSIONAL SUMMARY

Staff Data Engineer with 15+ years architecting and leading large-scale distributed data platforms at Warner Media, Microsoft, and JPMorgan Chase. Expert in building production streaming systems processing 10TB+/day with <100ms latency using Spark, Kafka, and cloud-native architectures. Track record of driving strategic cost optimization (\$700K+/year saved), leading cross-functional migrations affecting 15+ teams, and building reusable frameworks adopted across organizations. Deep expertise in Spark internals, real-time data architectures, and cloud data platform evolution (AWS, Databricks, Azure).

KEY ACHIEVEMENTS

- Led cross-functional data platform migrations impacting 15+ teams and 80+ production pipelines with zero downtime
- Built and scaled real-time streaming systems processing 10TB+/day with <100ms p99 latency serving 50M+ users
- Drove \$700K+ in annual cost savings through platform optimization initiatives
- Architected reusable frameworks reducing time-to-production by 85% for streaming use cases
- Established technical authority across 40+ engineers in Spark, Kafka, and cloud data platforms

EXPERIENCE

Senior Data Engineer — Warner Media – Discovery

London, UK · Feb 2022 – Present

Data platform modernization and real-time streaming for HBO Max AdTech

- Led enterprise migration from Snowflake to Databricks — framework and governance model adopted by 4 teams (15+ engineers, 80+ pipelines); 50% cost reduction (\$500K/yr) and 3x query performance.
- Architected the HBO Max AdTech data platform serving 50M+ daily users across 60+ countries; Lambda architecture (Databricks + Kafka/Spark Structured Streaming) processing 5TB/day at 99.95% SLA, cutting data latency from 4 hours to 15 minutes and enabling \$5M+ in targeted ad revenue.
- Designed the RTCM real-time streaming framework — 500K events/sec at <100ms p99, exactly-once, auto-scaling; adopted by 3 teams (12+ pipelines), cutting time-to-production from 6 weeks to 1.
- Drove a Spark performance program (data-skew salting, broadcast joins, AQE): 30% cluster cost cut (\$200K/yr), 40% faster jobs, 95% fewer OOM errors; trained 20+ engineers.

Senior Software Engineer — Microsoft

Bangalore, India · Jun 2021 – Dec 2021

Core contributor to the Azure Data Factory Spark execution engine

- Contributed to the Spark execution layer powering Azure Data Factory (4000+ enterprise clients, 100+ PB/month).
- Built a custom Spark Listener framework streaming 1M+ metrics/min to the ADF UI at <5s latency, cutting support tickets by 35%.
- Designed a high-cardinality metrics aggregation pipeline (HyperLogLog, T-Digest) holding sub-second queries at 10x peak load.

Senior Data Engineer — JPMorgan Chase & Co

Bangalore, India · Mar 2020 – Apr 2021

Real-time risk analytics platform for trading systems

- Led a near-real-time risk analytics platform processing 2M+ trades/day from 15+ sources using Spark Structured Streaming + Kafka with event-time aggregations, stream-stream joins, and exactly-once multi-sink writes.
- Cut risk-calculation latency from 30 minutes to under 2 — 15x faster decisions — supporting \$500B+ in daily trading volume.
- Migrated legacy DStream pipelines to Structured Streaming; an Avro schema registry gave 40% compression and safe schema evolution. Playbook adopted by 3 teams.

Technical Lead — Emids Technologies

Bangalore, India · May 2016 – Jun 2019

Enterprise healthcare data platform

- Architected the Enterprise Data Hub — a multi-tenant Cassandra healthcare data lake serving 8 insurers, 100M+ patient records, and 500GB/day ingestion; led a team of 6.
- Built the processing framework: schema validation with PII masking, dedup/cleanse, CDC, and graph-based (GraphX Connected Components) patient identity resolution at 98% accuracy — cutting an 8-hour job to 45 minutes.
- Designed a Java/Spring Boot microservices layer with Redis + Cassandra caching at <50ms p95 for 10K+ req/sec.

Senior Software Engineer — Amadeus Labs

Bangalore, India · Sep 2013 – Nov 2015

- Built a Spark/Flume pipeline for central logging — ingesting and parsing semi-structured logs and traces with automated anomaly detection.
- Deployed a scalable Flume-based collection system converting logs to Parquet with schema enforcement for downstream analytics.

TECHNICAL SKILLS

Core Strengths: Distributed Systems Architecture, Real-Time Streaming, Data Platform Modernization, Performance Optimization

Languages:	Java, Scala, Python, Golang
Data Engineering:	Apache Spark (Batch & Structured Streaming), Kafka, Flink, Delta Lake, Snowflake, Databricks, Airflow
Databases:	Cassandra, HBase, PostgreSQL, MySQL, Redis
Cloud Platforms:	AWS (S3, EMR, Kinesis, Redshift, Glue), Databricks (Unity Catalog, Delta Live Tables), Azure (ADF, Synapse)
DevOps & Tools:	Kubernetes, Docker, Terraform, Jenkins, GitLab CI/CD, Elasticsearch

FRAMEWORKS & PLATFORMS

RTCM Streaming Framework

[Spark Structured Streaming](#), [Kafka](#)

Reusable event-processing platform running 500K events/sec at <100ms p99 with exactly-once semantics and auto-scaling. Adopted by 3 teams, reducing time-to-production by 85%.

Databricks Migration Framework

[Databricks](#), [Delta Lake](#), [Unity Catalog](#)

Migration patterns and governance model used by 4 teams to move 80+ pipelines off Snowflake with zero downtime and a 50% cost reduction.

Spark Performance Toolkit

[Apache Spark](#), [AQE](#)

Best practices and automated tuning scripts deployed across 40+ engineers — codifying data-skew handling, broadcast joins, and AQE tuning into repeatable runbooks.

EDUCATION

B.Tech, Computer Science — [ICFAI University, Hyderabad](#)

2003 – 2007 · CGPA 8.09/10